Status	Finished
Started	Saturday, 24 October 2026, 4:33 PM
Completed	Saturday, 24 October 2026, 4:36 PM
Duration	3 mins 26 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search continues in the left sublist
c. The list is sorted
d. The search ends successfully ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search backtracks to find duplicate elements
c. The search starts over from the beginning
d. The search continues until the end of the list

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O👎
c. O(1)
d. O(log n) ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 90 and 100
- d. 94 and 99

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Linear search
- c. Binary search ✓
- d. Hash search

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By using a hash function
- c. By dividing the list length by two ✓
- d. By comparing each element sequentially

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Discovering
- b. Searching ✓
- c. Mining
- d. Finding

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the first element in the list
c. Compare the target element with the middle element in the list ✓
d. Sort the list

Question **9**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Binary ✓
c. Both (1) & (3)
d. Hash

Question **10**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O(1) ✓
d. O👎

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The lower middle element is chosen as the middle element ✗
b. The middle element is chosen randomly
c. The search stops
d. The higher middle element is chosen as the middle element

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and unsorted ✓
c. When the list is small and sorted
d. When the list is large and unsorted

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 10 ✓
c. 100
d. 1000

Question **14**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n-1
c. n/2
d. n ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the list contains duplicate elements
c. When the element is found in the list ✓
d. When the search algorithm finishes